First, the program initialises the microcontroller's registers, then it checks whether memory register is zero. If register r3 is zero, it goes for coin detection. Else, it proceeds to count update and display. Coin-counter register r3 is incremented by five after insertion of one coin. When the load switch is closed, port pin P3.1 goes low. Port pin P1.1 goes high to energise relay RL1. Port pin P3.2 goes low five times then display count number decrements by one.

EFY note. The source code and other relevant files of this article are included in this month's EFY-CD.

COIN.ASM

```
$mod51
;  p3.0              coin detect pulse
;  p3.1              power on switch
;  p3.2              monostable pulse(time
duration)sensed via transistor
;  p3.7              coin sensed LED
;  p1.0              monostable triggering
signal
;  p1.1              relay on or POWER BE-
ING CONSUMED LED indicator
;  p1.4 to p1.7    input to CD 4511 (
6.1,2,7)to display on 7seg
;  r0,r1             for delay
;  r2                count for 7 seg dis-
play
;  r3                count of 5 monostable
pulses (ASSUME Rs 1/1 MIN approx)
; ( r4               flag ON already trig-
gered
;  r5                flag    timer  already
triggered ) not used,for further devel-
opment
;  r6                count upto 5
;  r7                 count for 7 seg dis-
play left justified
        org 000h
        sjmp start
        org 040h
start:   ; -INITIALISATION START-
        mov r3,#000h ;count is 0
        mov r4,#00h ;flag reset
        mov r5,#000h ;mono on flag re-
set
        mov r2,#000h ;coin count 0
        mov  r6,#005h  ;counter set to
5
        mov r7,#00fh
        setb p3.0 ;no coin detected
        setb p3.2 ;mono output detected
set high
        clr   p3.7 ;coin detected LED
off
        clr   p1.1
; relay de-energised
                setb p1.0
; monostable not triggered
                clr   p1.4

                clr   p1.5
        ;  7 seg display 0
                clr   p1.6

                clr   p1.7

        ;-INITIALISATION OVER-
                acall delay
                 acall delay
                 acall delay
                 acall delay
                 acall delay
                 acall delay
                 acall delay
                 acall delay
                 acall delay
                 acall delay
tstcount:mov a,r3
        cjne a,#000h,tstpwrsw
        clr p1.1            ;if   r3=0
denergise relay
coindet: jnb p3.0,updtr3   ;  c o i n
sensed
        mov r4,#000h       ;flag reset
        ;-PUTTING  COUNT  ON  7  SEG
START-
        mov a,r2           ;no of coins
detected
        rl a
        rl a
        rl a                        ;
no of coins count in MS of  r2
        rl a
        mov b,a            ; copy in b
        mov a,p1
        anl     a,#00fh    ;extract LS
portin keep intact
        orl     a,b
;count ored in a
        mov   p1,a
        ;-PUTTING COUNT ON 7SEG OVER-
        sjmp tstcount
updtr3:  mov a,r3
        add a,#005h
        clr   c
        mov r3,a  ;5 added in r3
        mov a,r2  ; count no of coins
in r2
        inc a
        clr c
        cjne a,#10, max
        mov a,#9
        max:  mov r2,a
        acall coin                  ;
EVERY TIME COIN SENSED
        sjmp tstcount
tstpwrsw:jnb p3.1,swpwron
        clr  p1.1
        sjmp  coindet
swpwron: setb p1.1          ;relay on
        jnb p3.2,coindet  ;  is  delay
running ? if yes go and sense coin
        dec r3
        dec r6                      ;
reduce count from 5 set in r6
        mov a,r6
        cjne a,#000h,bypssr2
        dec r2                      ;1
subtracted  from r2 for every 5 in r3
        mov a,r2
        jz min
        rl a
        rl a
        rl a
        rl a
        clr  c
        mov b,a
        mov a,p1
        ani a,#00fh
        orl a,b
        mov p1,a
        mov r6,#005h       ;initial
count of 5 in r6
bypssr2: acall delay
        acall delay
        acall delay
        acall delay
        acall delay
        acall delay
        acall delay
        acall delay
trigr:   clr  p1.0
         acall delay                ;
mono triggered
        setb p1.0
        ajmp tstcount
        min:      mov p1,#01h
                  ajmp tstcount
                  ;-ROUTINES-
delay:          mov r0,#0c8h
loop2:   mov r1,#0fah
loop1:   nop
        nop
        nop
        djnz r1,loop1      ;loop1  ap-
prox 5 X 200=1msec
        djnz r0,loop2      ;loop2 250 X
1msec   =250 msec
        ret
coin:    setb  p3.7
        acall delay
        acall delay
        acall delay
        acall delay
        clr   p3.7
        acall delay
        acall delay
        acall delay
        acall delay
        ret
        end
```